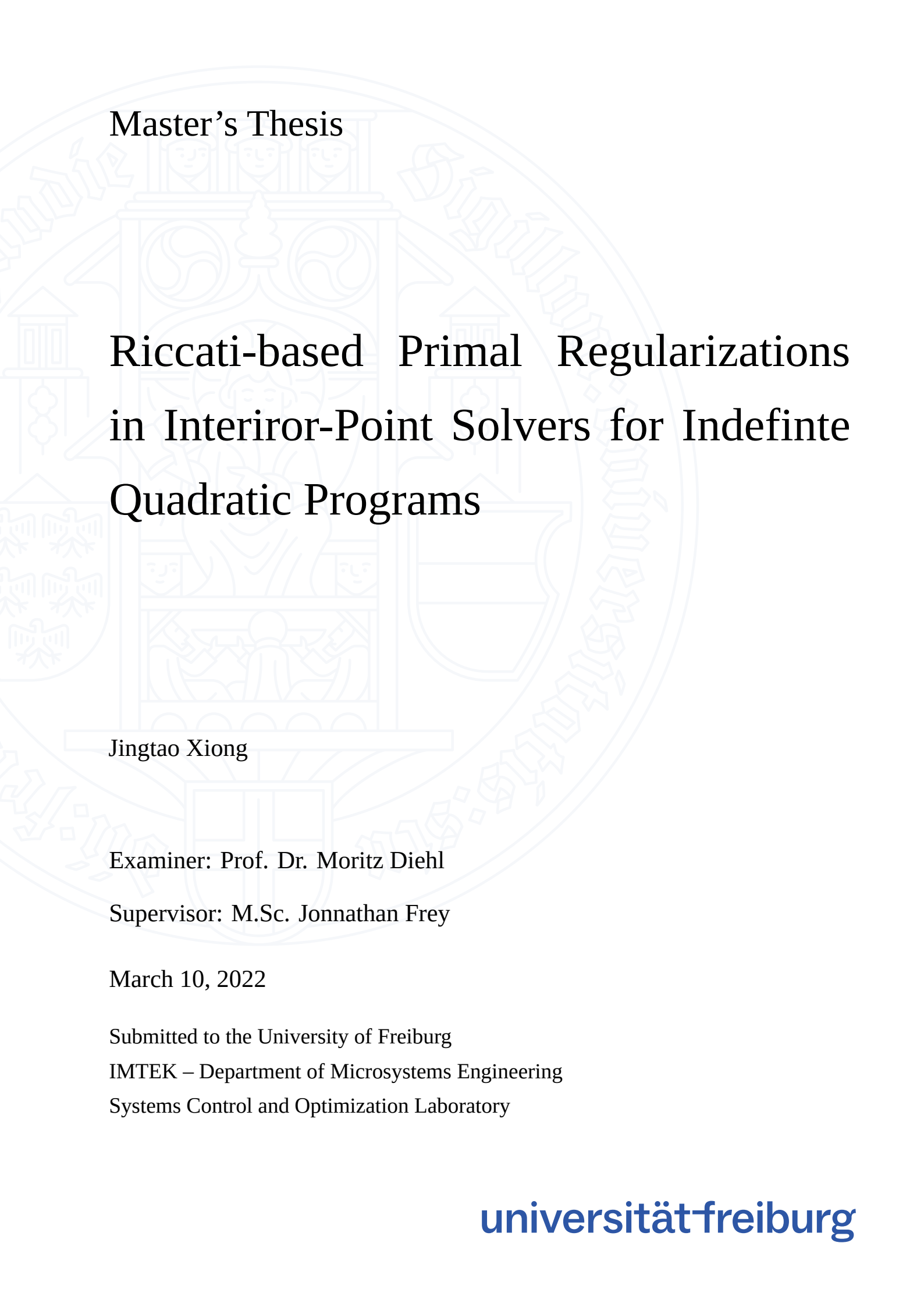




Master's Thesis

Riccati-based Primal Regularizations in Interior-Point Solvers for Indefinite Quadratic Programs

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Chapter 1

Primal Regularization for Indefinite Quadratic Programs

In this chapter, we turn our attention to the QPs with indefinite Hessian matrix, namely indefinite QPs. The chapter starts with a brief classification of the Hessian regularization in 1.1, then we introduce the concept of Riccati-based primal regularization in 1.2, the Riccati-based primal regularization method is consecutively introduced based on an analysis of failing on minimum perturbation in 1.3. We conclude the chapter with a proof of the asymptotic exactness of the proposed Riccati-based primal regularization method.

1.1 Primal regularization

In indefinite QPs, the regularization needs to be applied to the Hessian matrix to ensure a descent search direction in the interior point method. The resulting regularized reduced KKT system can be generally presented as:

$$\begin{bmatrix} \tilde{H} + E & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta \pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}_{\text{stat}}^i \\ r_{\text{eq}}^i \end{bmatrix}, \quad (1.1)$$

where E is the adjustment matrix for ensuring a positive definite Hessian. This process is referred to as primal regularization, and this section will briefly introduce its two types, distinguished by the target matrix, i.e. the matrix that the adjustment E is derived from and applied to, as the full Hessian regularization and the reduced Hessian regularization.

1.1.1 Regularization Strategies

We would like to start with the numerical techniques used by both full Hessian regularization and reduced Hessian regularization by regarding the calculation of adjustment matrix E .

There are several strategies for constructing the adjustment matrix E to ensure a positive definite Hessian, commonly, it can be computed from [1–3]

- Eigenvalue modification (EM): modifies the negative eigenvalues of $\tilde{H}$.
- Modified Cholesky factorization (MC): modifies the inertia of $\tilde{H}$.
- Levenberg-Marquardt (LM): adds a scaled identity matrix to $\tilde{H}$.

We will mainly apply four commonly used strategies for regularization in this work, two methods for EM (PROJECT, MIRROR) [4], one method for MC (GMW) [5], and one method for LM (Inertia Correction) [6].

Assuming a matrix A can be eigenvalue decomposed as $A = VD_EV^\top$ and Cholesky factorized as $A = LD_IL^\top$, where V is an orthogonal matrix of eigenvectors, L is a unit lower triangular matrix, D_E is a diagonal matrix of real eigenvalues, and D_I is a diagonal matrix whose signs give the inertia of A

The four methods can be summarized as follows:

- **PROJECT** (Minimal eigenvalue modification):

$$\text{project}(A, \epsilon) := V \max(\epsilon I, D_E) V^\top.$$

Where the max operator is applied component-wise. This approach clips any negative or near-zero eigenvalues to a small positive constant $\epsilon > 0$, which keeps a minimal adjustment to the original matrix for positive definiteness, thus the minimal Frobenius norm of the adjustment matrix E [1], but may lead to an increase in the condition number if the original eigenvalues were not close to zero.

- **MIRROR** (Absolute eigenvalue modification):

$$\text{mirror}(A, \epsilon) := V \max(\epsilon I, |D_E|) V^\top.$$

This method reflects negative eigenvalues to their positive counterparts. It preserves the magnitude of the curvature in all directions, which often provides better scaling, but may result in a large modification to the original matrix.

- **GMW** (Modified Cholesky factorization): Instead of a spectral decom-

position, GMW modifies the diagonal pivots on the fly during the $\text{LDL}^\top$ factorization. At the j -th step, given the trial pivot c_{jj} and the largest current off-diagonal $\theta_j = \max_{i>j} |c_{ij}|$, the pivot is bounded from below by

$$d_j^I = \max\left(\epsilon, |c_{jj}|, \frac{\theta_j^2}{\beta^2}\right), \quad \beta^2 = \max\left(\gamma, \frac{\xi}{\sqrt{n^2 - 1}}, \epsilon_M\right),$$

where $\gamma = \max_i |a_{ii}|$, $\xi = \max_{i \neq j} |a_{ij}|$, and ϵ_M is the machine precision. The term θ_j^2/β^2 bounds the entries of L and thus the condition number of the factor, while the implied correction is $e_j = d_j^I - c_{jj} \geq 0$, giving $A + E = LD_I L^\top$.

This avoids the cost of an eigendecomposition and still guarantees positive definiteness, but the resulting E is generally not minimal and depends on the variable ordering, so it tends to over-modify A compared with the eigenvalue-based methods.

- **Inertia Correction** (Levenberg-Marquardt):

$$\text{ic}(A, \Delta) := A + \Delta I, \quad \Delta \geq 0.$$

A uniform multiple of the identity is added, so every eigenvalue is shifted by the same amount, $\lambda_k(A + \Delta I) = \lambda_k(A) + \Delta$. In practice $\lambda_{\min}$ is not computed directly, and Δ is found by repeatedly attempting the $\text{LDL}^\top$ factorization and increasing Δ until the inertia is correct, i.e. until every diagonal entry of D_I is positive. The shift leaves the eigenvectors unchanged and is very cheap and robust, however, because it acts equally in all directions, it also inflates the already-positive eigenvalues and can degrade the scaling when only a few directions are indefinite.

1.1.2 Full Hessian regularization

The full hessian regularization aims at the positive definiteness of the Hessian matrix $\tilde{H}$. Commonly, in the optimal control structure, the full Hessian is regularized stage-wisely by operating on the block Hessian matrix $\tilde{H}_k$ at k stage,

$$\tilde{H}_k^{\text{reg}} = \begin{bmatrix} \tilde{Q}_k & \tilde{S}_k^\top \\ \tilde{S}_k & \tilde{R}_k \end{bmatrix} + E_k \succ 0, \quad k = 0, 1, \dots, N-1, \quad (1.2)$$

where E_k could be a diagonal matrix or a full matrix, depending on the regularization strategy mentioned in 1.1.1.

The full Hessian regularization is direct and usually simple to implement, but it can be too conservative without considering the definiteness of the reduced Hessian $\tilde{M}$, thus slow down the convergence.

1.1.3 Reduced Hessian regularization

The reduced Hessian regularization take the reduced Hessian matrix $\tilde{M}$ as the target matrix. According to the Theorem 1, which is mentioned in [4] (for proof, see [1]):

Theorem 1. *Assume that LICQ holds. The QP problem has a unique global optimum if and only if the reduced Hessian is positive definite:*

$$M = Z^\top H Z \succ 0. \quad (1.3)$$

The positive definiteness of the reduced Hessian is a necessary and sufficient condition for the QP admitting a unique global optimum. It is thus reasonable to derive the correction from the reduced Hessian $\tilde{M}$ instead of full Hessian $\tilde{H}$, which yields a less conservative regularization that still effective. In view of (1.2), a correction $E_k^r \succ 0$ derived from and applied to the reduced Hessian enters the KKT system as the adjustment matrix $E_k = Z_k E_k^r Z_k^\top$.

However, from definition, computing the reduced Hessian directly requires a null-space basis of the equality constraints, which can be achieved via condensing in optimal control problems [7], this step generally introduces extra computational overhead.

We thus propose a Riccati-based primal regularization method, which naturally obtains the reduced Hessian $\tilde{M}$ implicitly via Riccati recursion, and regularizes the reduced Hessian $\tilde{M}$ stage-wisely during the Riccati recursion to achieve a positive definite reduced Hessian $\tilde{M}$.

1.2 Riccati-based Primal Regularization

In this section, we will introduce the idea and concept of Riccati-based primal regularization, a stage-wise regularization method for the reduced Hessian $\tilde{M}$ based on the Riccati recursion in two sense: first, Riccati recursion is used for

detecting the indefiniteness of the reduced Hessian $\tilde{M}$, second, Riccati recursion is used for regularizing the reduced Hessian $\tilde{M}$ stage-wisely to ensure a positive definite reduced Hessian $\tilde{M}$.

1.2.1 Block Reduced Hessian From Riccati Recursion

As mentioned in [8], the Riccati recursion can be interpreted as a block factorization of the KKT system in (??). In this subsection, we will further show that the Riccati recursion can be regarded as the block $\text{LDL}^\top$ factorization of reduced Hessian $\tilde{M}^i$ in (??), which justifies the stage regularization on block reduced Hessian.

From the perspective of the Riccati recursion, the backward pass can be concluded as follows:

Algorithm 1 Backward Pass of Riccati Recursion

```

1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ 

2: for  $k = N - 1$  down to 0 do
3:    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$ 
4:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$ 
5:    $K_k \leftarrow -\tilde{M}_k^{-1} \Gamma_k$ 
6:    $k_k \leftarrow -\tilde{M}_k^{-1} \left( \tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) \right)$ 
7:    $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$ 
8:    $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
9: end for

```

We can find that $\tilde{M}_k = \tilde{R}_k + B_k^\top P_{k+1} B_k$ is exactly the block reduced Hessian obtained by Riccati recursion. In practice the inverse of $\tilde{M}_k$ is not computed directly; instead, the Cholesky factorization of $\tilde{M}_k$ is computed for efficiency.

From the perspective of the reduced Hessian $\tilde{M}^i$ in (??), eliminating the states through the dynamics leaves a dense reduced Hessian $\tilde{M}^i$ in the inputs $u_0, \dots, u_{N-1}$, whose density stems from the coupling of any two stages via the state transition $\Phi_{i,j}$, defined as:

$$\Phi_{i,j} = \begin{cases} \prod_{k=j}^{i-1} A_k = A_{i-1} A_{i-2} \cdots A_j, & \text{if } i > j, \\ I, & \text{if } i = j. \end{cases}$$

The backward pass in Algorithm 1 is exactly a block $\text{LDL}^\top$ factorization of this $\tilde{M}^i$, carried out without ever forming it explicitly. The connection is made

through the Schur complement: each cost-to-go matrix P_k is the Schur complement obtained by eliminating the states of the later stages, so that the pivot block $\tilde{M}_k$ is precisely the Schur complement of stage k in $\tilde{M}^i$ after all later stages have been eliminated. Consequently, the recursion $P_k = \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k$ is nothing but the successive formation of these Schur complements.

Collecting the pivots $\tilde{M}_{N-1}, \dots, \tilde{M}_0$ into the block-diagonal factor D and eliminating the stages in the same order as the backward pass, from stage $N-1$ down to stage 0, the block LDL[⊤] factorization of the reduced Hessian $\tilde{M}^i$ can be expressed as:

$$\tilde{M}^i = L \begin{bmatrix} \tilde{M}_{N-1} & 0 & \cdots & 0 \\ 0 & \tilde{M}_{N-2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \tilde{M}_0 \end{bmatrix} L^\top$$

where the accumulated back-substitution coefficients, built from the feedback gains K_k and the transitions $\Phi_{i,j}$, form the unit lower-triangular factor L :

$$L = \begin{bmatrix} I & 0 & \cdots & 0 & 0 \\ -B_{N-2}^\top K_{N-1}^\top & I & \cdots & 0 & 0 \\ -B_{N-3}^\top \Phi_{N-1,N-2}^\top K_{N-1}^\top & -B_{N-3}^\top K_{N-2}^\top & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -B_0^\top \Phi_{N-1,1}^\top K_{N-1}^\top & -B_0^\top \Phi_{N-2,1}^\top K_{N-2}^\top & \cdots & -B_0^\top K_1^\top & I \end{bmatrix}.$$

This establishes the connection between positive definiteness of the reduced Hessian $\tilde{M}^i$ and the positive definiteness of the block reduced Hessians $\tilde{M}_k$ obtained from Riccati recursion, which justifies the stage regularization on block reduced Hessian $\tilde{M}_k$.

1.2.2 Stage Regularization in Riccati Recursion

With this connection, we can regularize the reduced Hessian $\tilde{M}^i$ without explicitly computing it, but by stage-wisely applying the regularization to the block reduced Hessian $\tilde{M}_k^i$ during the Riccati recursion,

As mentioned in 1.2.1, the Cholesky factorization of $\tilde{M}_k^i$ is computed for efficiency, which requires the positive definiteness of $\tilde{M}_k^i$. This can be regarded as a detection of the indefiniteness of the reduced Hessian $\tilde{M}^i$. And when the Cholesky

factorization fails, which indicates that the reduced Hessian $\tilde{M}^i$ is indefinite, we can regularize the reduced Hessian block $\tilde{M}_k^i$ by adding a block adjust matrix E_k^i , i.e. $\tilde{M}_k^i + E_k^i$. The E_k^i can be computed from the strategies mentioned in 1.1.1. This gives a modification of the backward pass of Riccati recursion and can be concluded as follows:

Algorithm 2 Backward Pass of Modified Riccati Recursion

```

1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ ,  $\lambda > 0$ 

2: for  $k = N - 1$  down to 0 do
3:    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$ 
4:    $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
5:   if factorization fails then
6:      $\tilde{M}_k \leftarrow \tilde{M}_k + E_k$  ▷ Apply regularization
7:      $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
8:   end if
9:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$ 

10:   $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$ 
11:   $k_k \leftarrow -L_k^{-\top} L_k^{-1} (\tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}))$ 

12:   $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$ 
13:   $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
14: end for

```

Since the block reduced Hessians $\tilde{M}_k$ are exactly the pivots of the block $L^\top DL$ factorization of $\tilde{M}^i$, enforcing the positive definiteness of every $\tilde{M}_k$ during the backward pass is equivalent to regularizing the reduced Hessian $\tilde{M}^i$ itself, and thus recovers the SOSC in Theorem 1. The regularization is therefore applied only where it is needed while the structure-exploiting complexity of the Riccati recursion is preserved.

1.3 Improved Stage Regularization Strategy

In this section, we will analyze the case where the common-used minimum perturbation strategy in regularization fails in stage regularization, which causes a severe condition number issue in the preceding stages due to the propagation of the negative curvature through the stages. This motivates us to propose an improved stage regularization strategy that avoids minimum perturbation and

takes the propagation into account.

1.3.1 Failure of Minimum-Perturbation Regularization

In this subsection, we will show that the minimum perturbation strategy, i.e. the PROJECT [4], which is commonly the first choice for regularization with the least modification of the original problem, can cause condition number issue in the stage regularization by introducing a general example.

Suppose at the stage k , the reduced Hessian block $\tilde{M}_k$ is not positive definite. Without loss of generality, we can assume $\tilde{M}_k$ has only one negative eigenvalue λ_j , and the corresponding eigenvector is v_j . The minimum perturbation strategy is applied as:

$$E_k = (-\lambda_j + \epsilon)v_j v_j^\top, \quad (1.4)$$

where the ϵ is a small positive value to ensure the positive definiteness of the regularized reduced Hessian block.

By Sherman-Morrison formula, the inverse of the regularized reduced Hessian block can be expressed as:

$$\begin{aligned} (\tilde{M}_k + E_k)^{-1} &= \tilde{M}_k^{-1} + \frac{\lambda_j - \epsilon}{\epsilon \lambda_j} v_j v_j^\top \\ &= \tilde{M}_k^{-1} + \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) v_j v_j^\top, \end{aligned}$$

and the update of cost-to-go matrix P_k follows as:

$$\begin{aligned} P_k &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top K_k \\ &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) \Gamma_k^\top v_j v_j^\top \Gamma_k \\ &= P_k^{\text{unreg}} - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) \Gamma_k^\top v_j v_j^\top \Gamma_k, \end{aligned} \quad (1.5)$$

note that this negative rank-1 modification can already render P_k itself indefinite, independently of its downstream effect on $\tilde{M}_{k-1}$, whenever the perturbation magnitude exceeds the smallest eigenvalue of P_k^{unreg} . This can happen even when the perturbation on $\tilde{M}_k$ is moderate, provided that the negative-curvature direction v_j is strongly coupled to the state through Γ_k .

Consequently, the reduced Hessian block $\tilde{M}_{k-1}$ at the previous stage as:

$$\tilde{M}_{k-1} = \tilde{M}_{k-1}^{\text{unreg}} - \left(\frac{1}{\epsilon} - \frac{1}{\lambda_j}\right) z_j z_j^\top, \quad (1.6)$$

where $z_j = B_{k-1}^\top \Gamma_k^\top v_j$ is a column vector.

And the condition number of the regularized reduced Hessian block $\tilde{M}_{k-1}$ can be approximated as:

$$\kappa(\tilde{M}_{k-1}) \approx \frac{c(\epsilon + |\lambda_j|)}{\epsilon|\lambda_j|}, \quad (1.7)$$

where $c = \frac{\|z_j\|^2}{\lambda_{\min}(\tilde{M}_{k-1}^{\text{unreg}})}$ as an $O(1)$ constant.

Notably, as the ϵ is a small positive value, the presence of the negative rank-1 update $-\frac{1}{\epsilon} z_j z_j^\top$ will dominate, not only severely deteriorates its condition number, but also likely to make $\tilde{M}_{k-1}$ indefinite, which can further cause a domino effect through the stages.

1.3.2 Insights and Proposed Methods

This issue reveals two insights in stage regularization:

- The minimum modification objective [9] is not the foremost consideration for stage regularization, as it can cause severe condition number issue in the preceding stages. A trade-off between perturbation norm and condition number must be carefully treated.
- The cost-to-go matrix P_k must be incorporated into the regularization design, as the severity of propagation depends on the coupling structure between stages, which a purely local scheme cannot capture.

From the first insight, instead of the minimum perturbation strategy, an intuitive, but effective, approach is to use the absolute value of the negative eigenvalues or inertia as the perturbation magnitude, which can avoid the severe condition number issue in the preceding stages. When it comes to the eigenvalue modification strategy, this is the MIRROR strategy [4], and when it comes to the modified Cholesky factorization strategy, this is the Type-II strategy [9], of which GMW is a common representative.

From the second insight, we can incorporate the cost-to-go matrix P_k into the strategy design, which also can be achieved by adding an adjustment matrix $E_{k+1,P}^i$ to the cost-to-go matrix P_k directly, so that the propagation of the negative

curvature can be taken into account in the regularization design.

The resulting improved stage regularization strategy can be summarized as follows:

Algorithm 3 Backward Pass of Improved Modified Riccati Recursion

```

1: Initialize:  $P_N \leftarrow \tilde{Q}_N$ ,  $p_N \leftarrow \tilde{r}_{\text{stat},x}^N$ ,  $\lambda > 0$ 

2: for  $k = N - 1$  down to 0 do
3:    $P_{k+1}^{\text{reg}} \leftarrow P_{k+1}$  ▷ Default with no regularization
4:    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1}^{\text{reg}} B_k$ 
5:    $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
6:   if factorization fails then
7:      $E_{k+1,P} \leftarrow \mathcal{E}(P_{k+1})_\lambda - P_{k+1}$  ▷  $\mathcal{E}(\cdot)$ : MIRROR or Type-II strategy
8:      $P_{k+1}^{\text{reg}} \leftarrow P_{k+1} + E_{k+1,P}$  ▷ Regularize cost-to-go matrix
9:      $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1}^{\text{reg}} B_k$ 
10:     $E_{k,M} \leftarrow \mathcal{E}(\tilde{M}_k)_\lambda - \tilde{M}_k$ 
11:     $\tilde{M}_k \leftarrow \tilde{M}_k + E_{k,M}$  ▷ Regularize reduced Hessian block
12:     $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$ 
13:   end if
14:    $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1}^{\text{reg}} A_k$ 

15:    $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$ 
16:    $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left( \tilde{r}_{\text{stat},u}^k + B_k^\top (P_{k+1}^{\text{reg}} r_{\text{eq}}^k + p_{k+1}) \right)$ 

17:    $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1}^{\text{reg}} A_k + \Gamma_k^\top K_k$ 
18:    $p_k \leftarrow \tilde{r}_{\text{stat},x}^k + A_k^\top (P_{k+1}^{\text{reg}} r_{\text{eq}}^k + p_{k+1}) + \Gamma_k^\top k_k$ 
19: end for

```

In Algorithm 3, when the Cholesky factorization fails, the regularization acts at two levels. The term $E_{k+1,P}$ proactively convexifies the cost-to-go matrix that enters the current stage, so that the negative-curvature propagation identified in the second insight is accounted for in $\tilde{M}_k$, Γ_k , and P_k consistently. In contrast, $E_{k,M}$ acts as a local safeguard on the reduced Hessian block and is to ensure that the Cholesky factorization succeeds. Both perturbations are constructed from the absolute values of the negative eigenvalues (the mirror strategy) rather than the minimum-modification rule, which keeps the condition number of the regularized blocks bounded, as motivated by the first insight. When no factorization failure occurs, the recursion reduces to the standard Riccati backward pass.

1.4 Asymptotic Inactivity of the Regularization

In this section, we will prove that the reduced Hessian regularization via Riccati recursion would be asymptotically inactive, which means that the regularization will be inactive when the iterates are sufficiently close to the optimal solution (y^*, λ^*) , following a similar proof strategy to [10].

We will start with the regularized KKT formulation where the null space method is applied before the Schur complement to the (??), the resulting formulation is given as:

$$\begin{bmatrix} W & Z^\top G^\top \\ GZ & -\Lambda_i^{-1} T_i \end{bmatrix} \begin{bmatrix} \Delta y_z \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} r_{\text{stat, redu}}^i \\ \tilde{r}_{\text{ineq, redu}}^i \end{bmatrix}, \quad (1.8)$$

where

$$\begin{aligned} W &= M + E^i, \\ r_{\text{stat, redu}}^i &= Z^\top r_{\text{stat}}^i - Z^\top H Y \Delta y_y, \\ \tilde{r}_{\text{ineq, redu}}^i &= r_{\text{ineq}}^i - G Y \Delta y_y - \Lambda_i^{-1} r_{\text{comp}}^i, \end{aligned}$$

this formulation eliminates the equality constraints, recasting the optimal control structure into a general QP form, and yields a saddle-point system coupling the reduced primal variable Δy_z (with regularized reduced Hessian W) and the dual variable $\Delta \lambda$.

We will make the following assumptions for the proof:

Assumption 1. (y^*, λ^*) is a local (or global) minimizer for (??).

Assumption 2. Each local minimizer (y^*, λ^*) for (??) is isolated.

Assumption 3. Linear independence constraint qualification (LICQ) holds at the minimizer (y^*, λ^*) .

Assumption 4. Strict complementarity holds at the minimizer (y^*, λ^*) .

The Assumptions 1-3 imply the second order sufficient condition (SOSC) for local optimality, so we have the following lemma 2:

Lemma 2. Under Assumptions 1-3, for all $v \in C(y^*, \lambda^*) \setminus \{0\}$, it holds that $v^\top H v > 0$.

Proof. With Assumption 1, The second order necessary condition (SONC) for local optimality holds, which states that for all $v \in C(y^*, \lambda^*)$, we have $v^\top H v \geq 0$.

We proceed by contradiction. Suppose there exists $v \in C(y^*, \lambda^*) \setminus \{0\}$ such that $v^\top H v = 0$. Consider the point $y(\alpha) = y^* + \alpha v$ for $\alpha > 0$. Because the constraints are linear and v is in the critical cone, $y(\alpha)$ remains feasible for sufficiently small α .

Since the objective function of ?? is quadratic. Expanding $f(y(\alpha))$ yields:

$$f(y(\alpha)) = f(y^*) + \alpha \nabla f(y^*)^\top v + \frac{1}{2} \alpha^2 v^\top H v$$

As $\nabla f(y^*)^\top v = 0$, given the assumption that $v^\top H v = 0$, the equation reduces exactly to $f(y(\alpha)) = f(y^*)$. This implies that $y(\alpha)$ is also a local minimizer, which directly contradicts Assumption 2. Therefore, $v^\top H v > 0$ must hold. $\square$

Proposition 3. *Under Assumption 1-4, given the sequence $\{W^i\}$ constructed by the algorithm, as $\{(y^i, \lambda^i)\} \rightarrow (y^*, \lambda^*)$, $W^i = \tilde{M}^i = Z^\top \tilde{H}^i Z \succ 0$ for all i sufficiently large.*

Proof. Under Assumption 4, the critical cone $C(y^*, \lambda^*)$ reduces to the tangent space of the active constraints. Therefore, we have $C(y^*, \lambda^*) = \{v \mid Fv = 0, G_{\mathcal{A}}v = 0\}$.

For any $v \in \{v \mid \|v\| = 1\}$, we have the corresponding quadratic form of the reduced Hessian as:

$$\begin{aligned} v^\top \tilde{M}^i v &= d^\top (H + G^\top T_i^{-1} \Lambda_i G) d \\ &= d^\top H d + \sum_j \frac{\lambda_j}{s_j} (G_j d)^2 \\ &= d^\top H d + \sum_{j \in \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 + \sum_{j \notin \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 \end{aligned}$$

where $d = Zv \in \{d \mid Ad = 0\} \setminus \{0\}$.

We now proceed by contradiction, Assume $\exists v_i \in \{v \mid \|v\| = 1\}, v_i \rightarrow \bar{v} : v_i^\top \tilde{M}^i v_i \leq 0$. We can distinguish two cases:

For $G_{\mathcal{A}} \bar{d} = 0$, this indicates $d_i \in C(y^*, \lambda^*) \setminus \{0\}$. $d_i^\top H d_i > 0$ follows by Lemma 2. Rest of the terms are non-negative, so we have $\bar{v}^\top \tilde{M}^i \bar{v} > 0$, contradict with assumption.

For $G_{\mathcal{A}} \bar{d} \neq 0$, the term $\sum_{j \in \mathcal{A}} \frac{\lambda_j}{s_j} (G_j d)^2 \rightarrow \infty$, which dominates the other two terms, and thus $v_i^\top \tilde{M}^i v_i \rightarrow \infty > 0$, contradict with assumption.

Therefore, we can conclude that $W^i = \tilde{M}^i = Z^\top \tilde{H}^i Z \succ 0$ for all i sufficiently large, i.e., $E_i = 0$. $\square$

Proposition 3 shows the regularization is asymptotically inactive: once the iterates are close to (y^*, λ^*) , $E^i = 0$ and the regularized recursion coincides with the unregularized one. The solver thus recovers the exact Newton step and inherits the local convergence rate of the underlying interior-point method, without the regularization affecting its asymptotic behavior.

Chapter 2

Numerical Experiments

In this chapter, we evaluate the proposed regularization methods numerically and discuss the results. We begin in Section 2.1 with the two metrics used throughout the comparison, the iteration profile and the flop unit estimation, which measure convergence behavior and factorization cost respectively. Section 2.2 then describes the prototype solver on which all experiments are run and establishes that it reproduces the iteration behavior of HPIPm on convex QPs, so that any difference observed later can be attributed to the regularization rather than to the solver itself. Section 2.3 fixes the experiment setup, naming and classifying the regularization methods under comparison and stating the parameters shared by all of them. Section 2.4 presents the numerical comparison, first among the Riccati-based methods themselves and then against full Hessian regularization and inertia correction. Section ?? concludes the chapter by discussing what distinguishes the hard instances from the easy ones and why the full Hessian methods perform poorly.

2.1 Experiment Metrics

In this section, we will introduce the metrics used to evaluate the performance of different regularization strategies in our experiments. Since we care about both the convergence behavior and the factorization cost caused by different regularization methods, we will use two complementary metrics: iteration profile and flop unit estimation. The former one, iteration profile, is used to evaluate how many iterations each method takes to converge on a set of test instances, while the latter one, flop unit estimation, is approximated by the number of modified Cholesky factorization of the reduced Hessian, which is used to evaluate the factorization cost of each method.

2.1.1 Iteration Profile

Let $\mathcal{P}$ denote the set of test instances and $\mathcal{M}$ the set of candidate regularization strategies. For a method $m \in \mathcal{M}$ and an instance $p \in \mathcal{P}$, let $i_{m,p}$ be the number of outer iterations at which m satisfies the convergence criterion on p . A run that reaches the iteration cap $i_{\max}$ without converging, or that stalls early, is treated as unsolved and we set $i_{m,p} = \infty$.

The *iteration profile* of method m is

$$\pi_m(k) = \left| \{p \in \mathcal{P} : i_{m,p} \leq k\} \right|, \quad k = 1, \dots, i_{\max}, \quad (2.1)$$

i.e. the number of instances that m solves within a budget of k iterations. Equivalently, $\pi_m(k)/|\mathcal{P}|$ is the empirical cumulative distribution of the solve-iteration count over solved instances. By construction π_m is non-decreasing in k , and its value at the cap, $\pi_m(i_{\max})$, equals the total number of instances solved by m under the budget.

The profile is read along two axes. The rate at which $\pi_m(k)$ rises with k reflects the *efficiency* of method m : a curve that climbs steeply at small k solves most instances with few iterations. The height of the plateau, $\pi_m(i_{\max})$, reflects the *robustness* of the method: it is the fraction of the test set solved within the prescribed budget. Because every method is evaluated under the same cap $i_{\max}$ and the same convergence criterion, the curves are directly comparable across $\mathcal{M}$.

This metric gives a summary of how reliably and how quickly each strategy converges across the test set.

2.1.2 Flop Unit Estimation

Iteration counts alone do not reflect the work performed per iteration, which differs substantially across regularization strategies: an eigenvalue-based correction and a modified-Cholesky correction may reach convergence in a comparable number of iterations while incurring very different arithmetic cost. To compare methods on a common scale, we express all dense linear-algebra work in a single flop unit.

We take one modified Cholesky factorization of the block reduced Hessian as the unit. For a control dimension n_u , factorizing an $n_u \times n_u$ matrix costs approximately $\frac{1}{3}n_u^3$ flops, hence

$$\text{unit} \approx \frac{1}{3}n_u^3. \quad (2.2)$$

The modified factorization differs from the plain one only by lower-order diagonal updates, so to leading order it also costs one unit.

The remaining primitives are expressed relative to this unit. A symmetric eigenvalue decomposition of the same block costs about $9n_u^3$ flops [11], i.e.

$$\frac{9n_u^3}{\frac{1}{3}n_u^3} = 27 \text{ units.} \quad (2.3)$$

With this unit defined, the arithmetic cost of a solve, including the regularization applied at each iteration, can be expressed on a common scale and accumulated over the iteration profile.

2.2 Prototype Solver Implementation

To unify the comparison, we implemented a prototype solver capable of accommodating all regularization methods under consideration. The Riccati-based primal-dual interior-point solver is implemented in Python, using HPIPM, the state-of-the-art Riccati-based IPM solver for OCP QPs, as a reference. The prototype was designed to reproduce its iteration behavior on convex QPs, where the two solvers should therefore agree.

One implementation detail deserves mention, as it departs from the plain FTB rule stated in Section ???. Rather than retracting the maximal step by a fixed fraction η , the primal and dual step sizes α^{prim} and α^{dual} are each obtained from their respective maximal step through the heuristic

$$\alpha = \begin{cases} \alpha_{\max} \left[(1 - \alpha_{\max}) \kappa_{\eta} + \alpha_{\max} \hat{\eta} \right], & \alpha_{\max} < 1, \\ \alpha_{\max}, & \alpha_{\max} = 1, \end{cases} \quad (2.4)$$

where $\alpha_{\max} \in (0, 1]$ denotes the maximal step length for ensuring the positivity of the slack and dual variables, $\kappa_{\eta} = 0.99$ and $\hat{\eta} = 1 - 10^{-7}$. The bracketed factor interpolates linearly between the conservative constant κ_{η} and the near-unit value $\hat{\eta}$, with $\alpha_{\max}$ itself as the interpolation weight. The separate case $\alpha_{\max} = 1$ is needed, otherwise the first branch would scale an unblocked full step by $\hat{\eta}$ and prevent it from ever being taken exactly, thus affecting the local quadratic convergence rate of IPM.

For the comparison to be meaningful, We align the prototype solver with HPIPM in terms of the algorithmic choices. We therefore define a set of HPIPM

settings, referred to in the following as the CUSTOMIZED mode, the values are listed in Table 2.1.

Table 2.1: Detailed settings for the defined CUSTOMIZED mode.

Parameter	Value	Parameter	Value
mu0	1e1	lam_min	1e-12
alpha_min	1e-12	t_min	1e-12
res_g_max	1e-6	tau_min	1e-12
res_b_max	1e-8	lam0_min	1e-12
res_d_max	1e-8	t0_min	1e-12
res_m_max	1e-6	warm_start	0
dual_gap_max	1e15	abs_form	0
iter_max	30	comp_dual_sol_eq	1
stat_max	30	comp_res_exit	1
pred_corr	1	comp_res_pred	1
cond_pred_corr	1	split_step	1
itref_pred_max	0	m_safe	0.5
itref_corr_max	0	var_init_scheme	0
reg_prim	0.0	t_lam_min	2
square_root_alg	0	t0_init	2
lq_fact	0	update_fact_exit	1

As one can notice, the CUSTOMIZED mode largely coincides with the SPEED mode of the original HPIPM, differing in two respects. First, the square-root algorithm is disabled (`square_root_alg=0`), so that the Riccati recursion is carried out directly rather than on Cholesky factors. Second, the primal and dual step sizes are computed separately (`split_step=1`) instead of sharing a single step length. In both modes no iterative refinement is applied to the predictor or corrector steps (`itref_pred_max=itref_corr_max=0`) and no default primal regularization is added (`reg_prim=0`), while Mehrotra’s predictor–corrector is retained and applied conditionally (`pred_corr=cond_pred_corr=1`).

To verify that the two implementations do agree, we collected 40 convex QP subproblems, five from each of the 8 OCPs in the benchmark, and compared the iteration behavior of the prototype solver against that of CUSTOMIZED-mode HPIPM. The resulting iteration profiles are shown in Figure 2.1.

Both solvers solve all 40 instances in the exact same number of iterations for each instance, resulting in performance profiles that coincide across the whole budget range, from 4 up to 22 iterations. On convex QPs the prototype solver therefore reproduces the iteration behavior of HPIPM, which verifies the prototype

as a reliable baseline and allows us attribute the differences observed in the experiments to the regularization rather than to the solver.

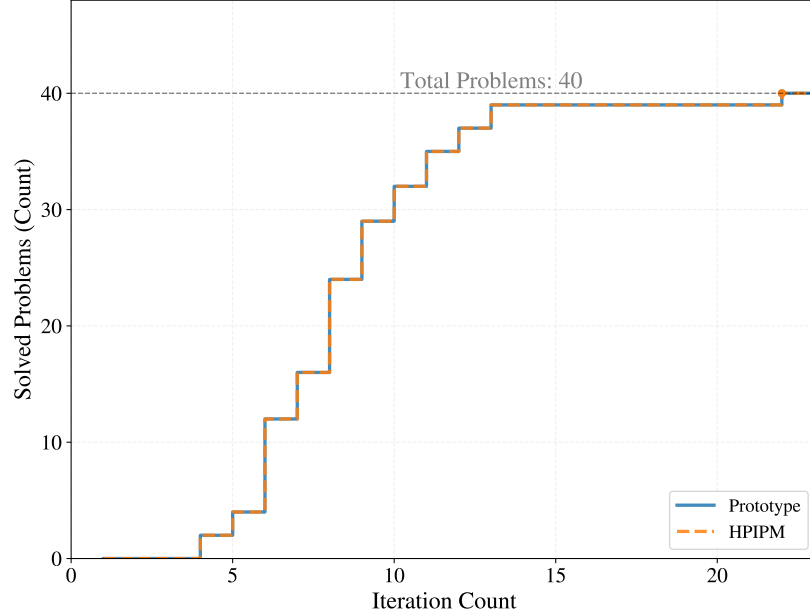


Figure 2.1: Iteration profiles on 40 convex QPs. The prototype solver is compared with HPIPm, showing identical iteration behavior.

2.3 Experiment Setup

In this section, the settings preparation for the experiment are introduced. First, based on the previously discussed methods, we will name and classify the regularization methods considered in the experiment, then we will introduce the general setup of the experiments, including the convergence criterion and step-size control strategy.

2.3.1 Experimented Regularizations

Before turning to the results, we fix the set of regularization methods entering the comparison, together with the naming convention used throughout this chapter.

Following the classification in Section 1.1, the methods are grouped by their *target*, that is, by the matrix whose indefiniteness the correction is derived from and to which the correction is directly applied, rather than by how the correction itself is computed. This yields three families: full Hessian regularization, Riccati-based regularization, and inertia correction. The numerical strategies

of Section 1.1.1, namely spectral projection (**PROJECT**), eigenvalue mirroring (**MIRROR**) and the Gill–Murray–Wright modified Cholesky factorization (**GMW**), are largely independent of the matrix they act on and can therefore be paired with each target.

Every method is accordingly named in the form **TARGET_STRATEGY**, so that both factors can be read off directly from the label, and the comparison can be carried out either across targets at a fixed strategy or across strategies at a fixed target.

Full Hessian Regularization The first family follows Section 1.1.2 and enforces positive definiteness of the full Hessian $\tilde{H}$ in the KKT system. The regularization is also triggered by the failure of the Cholesky factorization in Riccati recursion, the regularized full Hessian is obtained by stage-wisely regularizing each block $\tilde{H}_k$ and the Riccati recursion is re-attempted with it. The three variants differ only in how the block adjustment E_k is formed:

- **H_PROJECT**: the negative and near-zero eigenvalues of each $\tilde{H}_k$ are clipped to ϵ , which gives the smallest perturbation in Frobenius norm that restores positive definiteness.
- **H_MIRROR**: the eigenvalues of each $\tilde{H}_k$ are replaced by their absolute values (floored at ϵ), which preserves the magnitude of the curvature along every eigendirection at the price of a larger modification.
- **H_GMW**: the diagonal pivots of each $\tilde{H}_k$ are bounded from below during the $\text{LDL}^\top$ factorization, which avoids an eigendecomposition altogether and keeps the entries of the triangular factor bounded, but yields a non-minimal, ordering-dependent perturbation.

Riccati-based Regularization The second family is the one proposed in Section 1.2 and acts on the reduced Hessian $\tilde{M}$. The algorithms are given in Algorithms 2 and 3, where the regularization is applied stage-wise to the block reduced Hessian $\tilde{M}_k$ during the backward pass, when the failure of the Cholesky factorization of $\tilde{M}_k$ occurs.

The two algorithms differ in which matrices of the backward pass the adjustment is derived from and applied to, giving two targets within this family:

- **M**: only the block reduced Hessian $\tilde{M}_k$ is corrected, as in Algorithm 2. The correction is purely local and serves as a minimal safeguard for the factorization.

- **PM**: the cost-to-go matrix P_{k+1} entering the stage is corrected as well, as in Algorithm 3 introduced in this work, so that the propagation of negative curvature analysed in Section 1.3 is accounted for consistently in $\tilde{M}_k$, Γ_k and P_k .

Pairing each target with the three strategies gives the six methods compared in this chapter. Each method accordingly answers a specific question in the comparison:

- **M_PROJECT**: the reference point of the family. A minimum-norm correction applied locally is the naive and cheapest way to enforce positive definiteness, and is the configuration that the analysis of Section 1.3 predicts to be the most exposed to condition-number growth.
- **M_MIRROR**: whether abandoning the minimum-perturbation rule is by itself sufficient, with the propagation left untreated.
- **M_GMW**: the same question as **M_MIRROR**, but with the correction on the inertia of the factorization rather than on the spectrum, which indicates whether the cheaper Type-II correction can stand in for mirroring.
- **PM_PROJECT**: conversely, whether regularizing the cost-to-go matrix helps even when the minimum-perturbation rule is retained, so that this case can be discussed rather than merely argued for.
- **PM_MIRROR**: both insights combined, and thus the method the analysis of Section 1.3 recommends.
- **PM_GMW**: combines both insights with a cheaper factorization-based correction to explore the trade-off between robustness and decomposition cost compared to **PM_MIRROR**.

Inertia Correction The third family is the Levenberg–Marquardt style inertia correction used in IPOPT [6], which we denote by **INERTIA_LM** and take as the practical reference point for the comparison.

In this strategy, a scalar multiple of the identity $\delta_w I$ is added stage-wise to the full Hessian, i.e. each full Hessian block $\tilde{H}_k$ is replaced by $\tilde{H}_k + \delta_w I$, and the Riccati recursion is repeated with increasing δ_w until it succeeds. Only the primal regularization from IPOPT is retained, as dual regularization is not the primary focus of this work.

Notably, the method sits between the two preceding families, as it has no distinct target matrix in the above sense. Being a uniform multiple of the identity, the shift is not tied to any particular direction, so the perturbation remains a

uniform shift along every direction for both the full and the reduced Hessian. Accordingly, as $\delta_w I$ is applied uniformly to Q_k and R_k in full Hessian blocks $\tilde{H}_k$, it enters the Riccati recursion as the same additive term on the reduced Hessian blocks $\tilde{M}_k$ and P_k .

The trial sequence follows IPOPT as well: the unmodified matrix is attempted first ($\delta_{\text{LM}} = 0$), the first correction of a solve starts from δ_w^0 and is then scaled by the accelerated factor $\bar{\kappa}_w^+$, subsequent corrections are scaled by κ_w^+ until the Riccati recursion succeeds or the threshold δ_w^{max} is exceeded, and a successful value is stored and reduced by κ_w^- as the starting point of the next iteration. The parameters are set to the IPOPT defaults and are listed in Table 2.2.

Table 2.2: Inertia Correction (IC) Parameters

Symbol	Value	Description
δ_w^0	10^{-4}	Initial Hessian perturbation parameter
δ_w^{min}	10^{-16}	Minimum allowable regularization parameter
δ_w^{max}	10^{16}	Maximum allowable regularization threshold
κ_w^-	$1/3$	Factor to decrease perturbation upon success
κ_w^+	8	Factor to increase perturbation after inertia failure
$\bar{\kappa}_w^+$	10^2	Accelerated scaling factor for severe indefiniteness
δ_{LM}	0.0	Initial LM shift (IC-1: attempt unmodified matrix first)

Taken together, the three families span ten methods, which share the same solver, the same benchmark and the same parameters introduced in the next subsection,

2.3.2 General setup

All ten methods are run under the same conditions. The barrier parameter is updated by Mehrotra probing (`pred_corr`= 0) and the step size is controlled by the FTB rule followed by the funnel line search, while the remaining settings are collected in Table 2.3.

A run terminates either when all four residuals fall below their tolerances or when the iteration budget i_{max} is exhausted, in which case the instance counts as unsolved.

Table 2.3: Settings shared by all experiments.

Symbol	Value	Role
r_{stat}	10^{-6}	Tolerance on the stationary residual
r_{eq}	10^{-8}	Tolerance on the equality residual
r_{ineq}	10^{-8}	Tolerance on the inequality residual
r_{comp}	10^{-6}	Tolerance on the complementarity residual
i_{max}	120	Iteration budget, beyond which the instance counts as unsolved
$n_{\mathcal{P}}$	175	Nonconvex QP instances in the test set

The funnel line search brings a few parameters of its own, which are listed in Table 2.4 and kept fixed across all runs.

Table 2.4: Parameters of the funnel line search.

Symbol	Value	Role
β	0.5	Backtracking factor, $\alpha_p \leftarrow \beta \alpha_p$
n_{max}	10	Maximum backtracking iterations
θ_{max}^0	100	Initial max funnel bound
κ_s	10^{-1}	Switching constant, $\alpha_p(-\rho) > \kappa_s \theta^2$
γ	10^{-4}	Armijo constant for f-step, $f^+ \leq f + \gamma \alpha_p \rho$
κ_θ	0.999	h-step acceptance factor, $\theta^+ \leq \kappa_\theta \theta_{\text{max}}$
κ_u	0.5	Funnel-update weight, $(1 - \kappa_u)\theta_{\text{max}} + \kappa_u \theta^+$

2.4 Numerical Comparison of Primal Regularizations

In this section, we report the numerical results of the ten regularization methods on the nonconvex QPs of the benchmark. The comparison proceeds in three stages, each narrowing the field. We first compare the six Riccati-based methods against one another, in order to identify the strongest representatives of the family. These are then set against the full Hessian regularizations, and finally against inertia correction, which serves as the practical reference point.

2.4.1 Comparison among Riccati-based regularizations

We first compare the proposed Riccati-based regularization methods, which are the main focus of this work. As a baseline we also include the unregularized recursion (`NO_REGULARIZE`).

The iteration profile in Figure 2.2 reports, for each method, the number of instances solved within a given iteration budget, where two trends stand out. First, each **PM** dominates its counterpart **M** across the whole budget range, second, among the regularization strategies, **MIRROR** and **GMW** are clearly more effective than **PROJECT**. These trends validate the insights mentioned in Section 1.3. Among all the methods, **PM_MIRROR** tops the profile, solving 171 of the 175 instances, with **PM_GMW** close behind at 168.

Besides, it is interesting to note that the unregularized recursion can also solve a non-negligible fraction of the test set (31 of the 175 instances), which can be explained by the curvature contribution of the constraint that help to make the reduced Hessian positive definite in some instances.

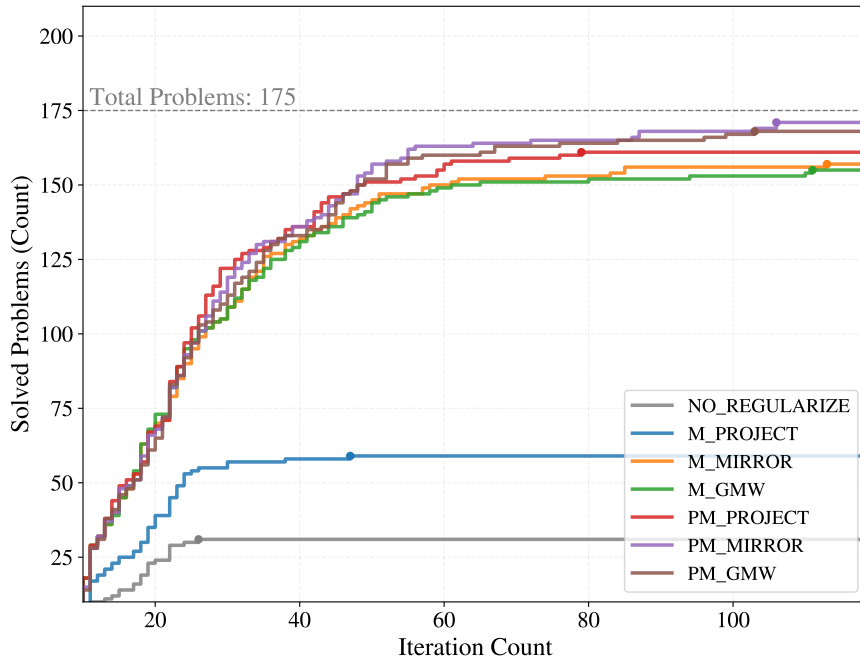


Figure 2.2: Iteration profile of the six Riccati-based methods over the 175-instance test set: number of instances solved within a given iteration budget.

Figure 2.3 complements this picture by showing the distribution of the worst-case decomposition cost in flop estimation, annotated with the maximum observed cost and the number of failures for each method. Since only the successful

instances and maximal iteration instances are counted, the **M_PROJECT** will be excluded from the analysis, due to the frequent failing on 116 of the 175 instances.

In general, the **PM** incur a higher and more dispersed cost, reflecting the extra factorization work introduced by the cost-to-go term, yet they are markedly more robust and thus the focus of the comparison. In Figure 2.3, **PM_MIRROR** is the most robust method with only 4 unsolved problems, and **PM_GMW** follows with 7 while keeping its worst-case cost more than an order of magnitude below **PM_MIRROR** (6.5×10^4 versus 9.4×10^5 flops) and nearly a factor of 40 below **PM_PROJECT** (2.4×10^6). Notably, though the **PM_PROJECT** and **PM_MIRROR** belong to the same class of regularization strategy, the worst-case cost of **PM_PROJECT** is higher than that of **PM_MIRROR**, which can be explained by the issue of inadequate regularization in the minimum perturbation strategy, so more frequency is needed in the **PM_PROJECT** case.

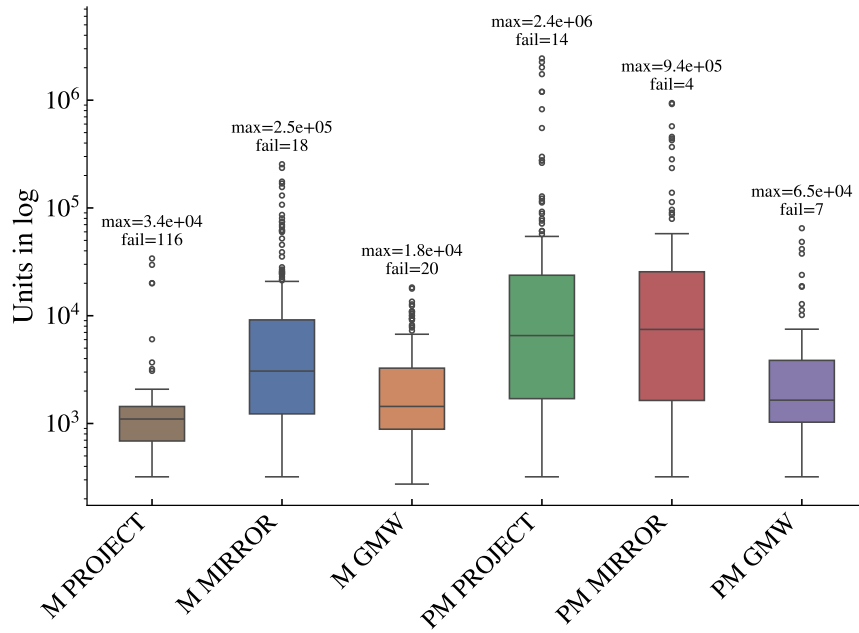


Figure 2.3: Worst-case decomposition cost of the six Riccati-based methods (flops, log scale). Annotations report the maximum observed cost and the number of unsolved instances.

Taken together, two figures first reveal a robustness ordering, **PM_MIRROR** > **PM_GMW** > **PM_PROJECT** > {**M_MIRROR**, **M_GMW**} > **M_PROJECT**, in which every **PM** outranks every plain **M**. If reliability is the priority, **PM_MIRROR** is the safest choice, as it leaves the fewest instances unsolved. If a balance between reliability and computational cost is preferred, **PM_GMW** is the more attractive

option: it recovers almost all of the robustness of **PM_MIRROR** (7 unsolved instances against 4) at a worst-case cost roughly $15\times$ smaller. We will therefore focus on **PM_MIRROR** and **PM_GMW** in the following comparisons with other regularization families.

2.4.2 Comparison with Full Hessian Regularizations

Having identified **PM_MIRROR** and **PM_GMW** as the strongest Riccati-based methods, we now compare them against the full Hessian regularizations.

The iteration profile in Figure 2.4 shows the two Riccati-based methods sitting well above every full-Hessian variant across the whole iteration budget. **PM_MIRROR** and **PM_GMW** leave only 4 and 7 instances unsolved, whereas each **H** method leaves a substantial number of instances unsolved even at the largest budget. This suggests that regularizing the full Hessian does not recover the reliability or efficiency obtained by regularizing within the Riccati recursion.

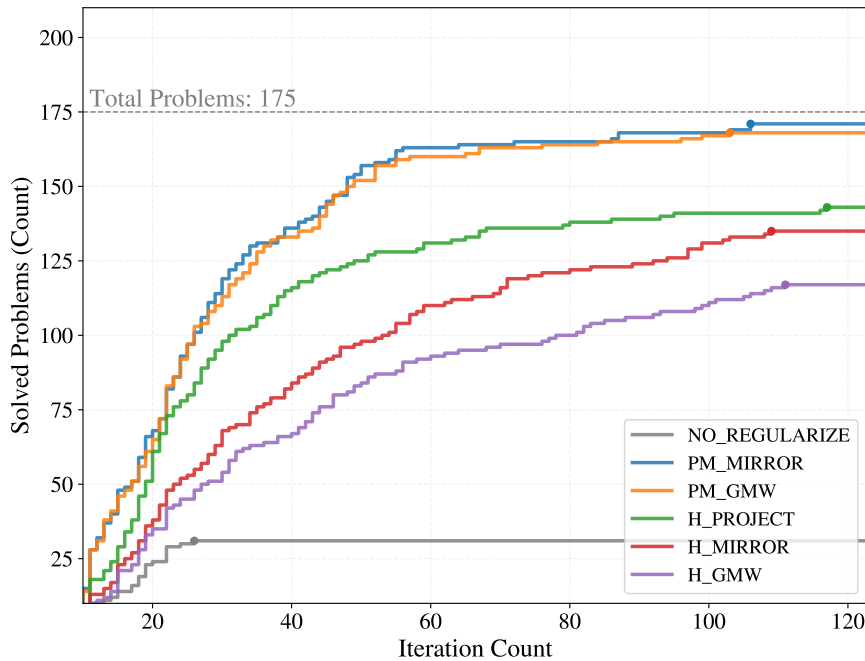


Figure 2.4: Iteration profile of the two strongest Riccati-based methods (**PM_MIRROR**, **PM_GMW**) against the full Hessian regularizations (**H_PROJECT**, **H_MIRROR**, **H_GMW**).

In Figure 2.5, the gap between Riccati-based methods and Full Hessian methods is even more pronounced, and it runs in the Riccati-based methods' favour on *both* axes. In robustness, **PM_MIRROR** fails on only 4 instances and **PM_GMW**

on 7, while the full-Hessian methods fail on 32 (H_PROJECT), 40 (H_MIRROR), and 58 (H_GMW) instances. In cost, the full-Hessian variants are one to several orders of magnitude more expensive: their worst-case costs reach 1.5×10^7 flops for H_PROJECT and 2×10^7 for H_MIRROR, and even the cheapest full-Hessian option, H_GMW, has a median cost roughly two orders above that of PM_GMW, yet it is also the least robust of the methods compared here. The full Hessian methods thus are considered neither speed nor reliability relative to the Riccati-based approach.

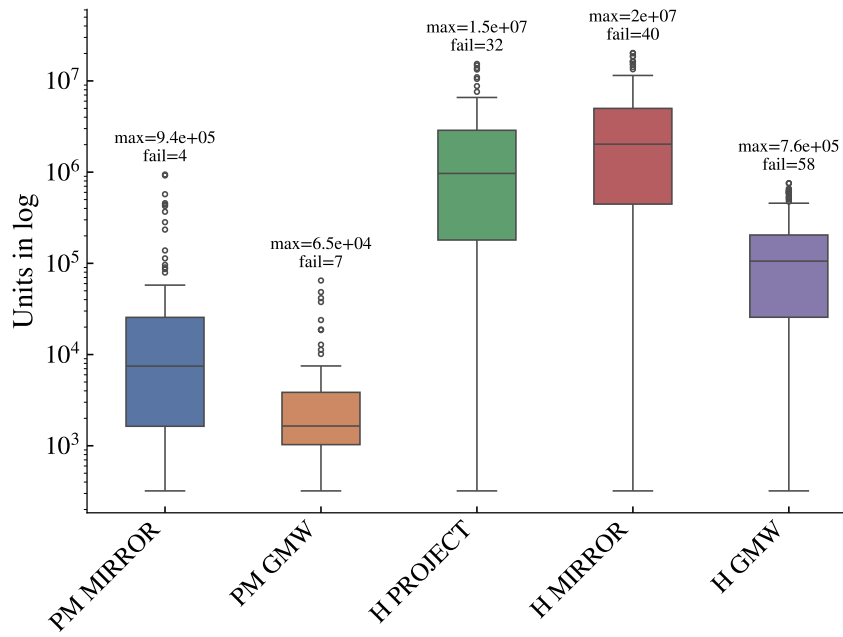


Figure 2.5: Worst-case decomposition cost of the two strongest Riccati-based methods against the full Hessian regularizations (flops, log scale). Annotations as in Figure 2.3.

Unlike the within-family comparison, where robustness and cost traded off against one another, the Riccati-based methods are simultaneously more robust and cheaper than every full-Hessian counterpart in Figure 2.5. This supports the central premise of the approach, namely that regularizing the reduced Hessian by exploiting the stagewise structure of the problem is preferable to regularizing the full Hessian. Consequently, PM_MIRROR and PM_GMW remain the recommended methods, PM_MIRROR when reliability is paramount, and PM_GMW when a lower and more predictable decomposition cost is desired at the price of a handful of unsolved instances.

2.4.3 Comparison with Inertia Correction

Finally, we compare the recommended Riccati-based methods against inertia correction (INERTIA_LM), which is widely used in practice and serves as a reference point for the comparison.

The iteration profiles in Figure 2.6 are close for the three regularized methods, with PM_MIRROR and PM_GMW solving the most instances (171 and 168) and INERTIA_LM tracking them closely across the budget range (160). However, the PM_MIRROR and PM_GMW generally sits slightly above INERTIA_LM, indicating less iteration is needed to solve the same number of instances.

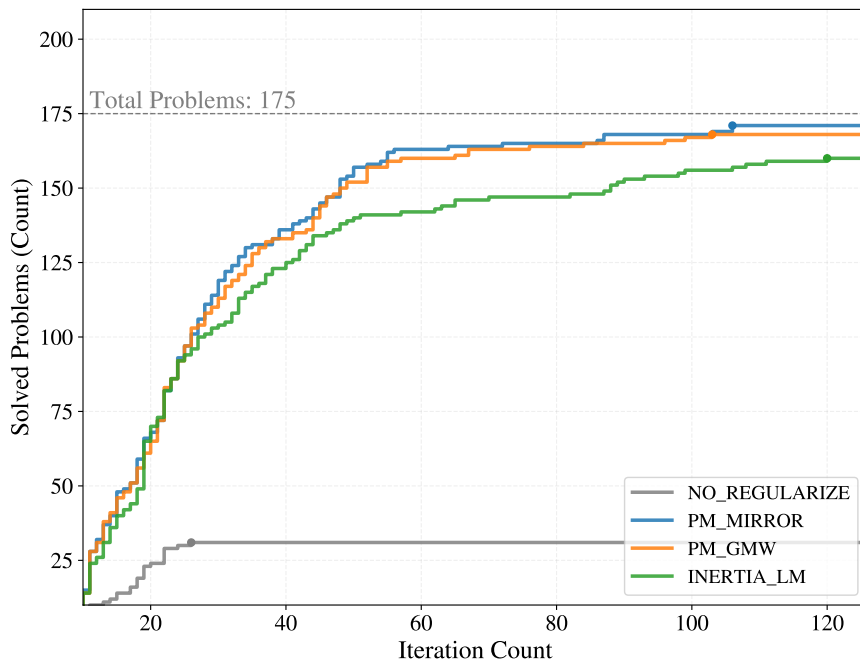


Figure 2.6: Iteration profile of PM_MIRROR and PM_GMW against inertia correction (INERTIA_LM).

The decomposition-cost distribution in Figure 2.7 draws the distinction more finely. Inertia correction is the cheapest of the three in the worst case, with a maximum cost of 2.4×10^4 flops, below both PM_GMW (6.5×10^4) and PM_MIRROR (9.4×10^5). In terms of median cost, however, INERTIA_LM and PM_GMW are essentially on par, with PM_GMW marginally lower. Their robustness differs more clearly, however: INERTIA_LM fails on 15 instances, against 7 for PM_GMW and 4 for PM_MIRROR.

The comparison thus no longer reduces to an even trade-off between INERTIA_LM and PM_GMW. INERTIA_LM retains a single advantage, the lowest

worst-case cost, but a narrow one: a factor of roughly 2.7 below **PM_GMW**, against which **PM_GMW** is in turn slightly cheaper at every quartile of the distribution and solves eight more instances.

Inertia correction is therefore preferable only where a hard bound on the per-decomposition cost outweighs the number of solved instances. **PM_MIRROR**, meanwhile, continues to stand apart as the most robust method of the three, and strictly so: the four instances it leaves unsolved are also unsolved by **PM_GMW**, so its additional robustness costs nothing in reliability elsewhere, albeit at a substantially higher worst-case cost.

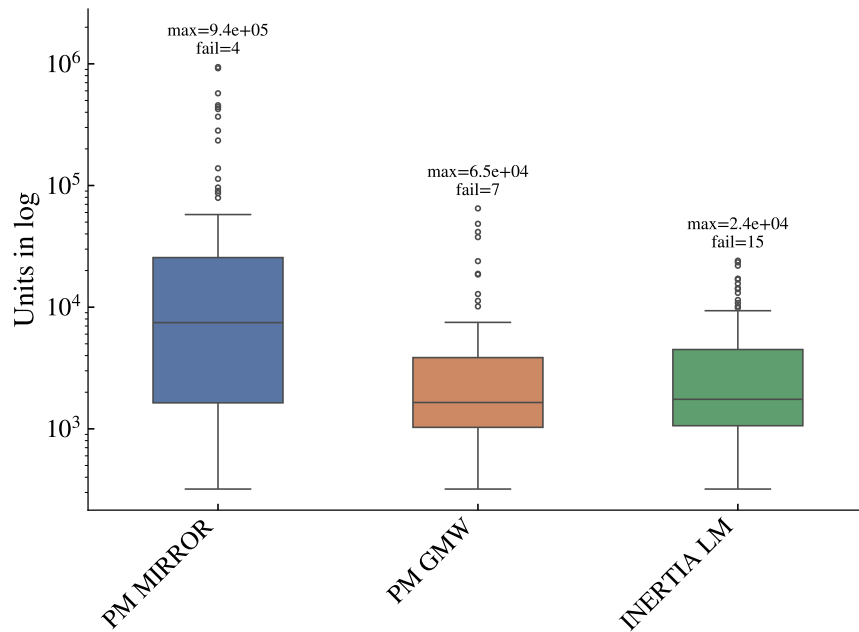


Figure 2.7: Worst-case decomposition cost of **PM_MIRROR** and **PM_GMW** against inertia correction (flops, log scale). Annotations as in Figure 2.3.

2.5 Discussion

2.5.1 What kind problem is hard or easy

2.5.2 Why Full Hessian fail and bad

2.5.3 Ablation on funnel line search

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